

1. Descriptive Statistics

1.1 Measures of Central Tendency

Q1: What is the measure of central tendency that represents the sum of all values divided by the number of values?

- a) Median
- b) Mode
- c) Mean
- d) Range

Q2: The median is:

- a) The most frequently occurring value in a data set
- b) The sum of all values divided by the number of values
- c) The middle value when the data is arranged in ascending or descending order
- d) The difference between the highest and lowest values

Q3: Which measure of central tendency is most affected by extreme values?

- a) Median
- b) Mode
- c) Mean
- d) Range

1.2 Measures of Variability

Q4: The difference between the highest and lowest values in a dataset is called:

- a) Range
- b) Variance
- c) Standard deviation
- d) Median

Q5: What does a high standard deviation indicate in a dataset?

- a) The data points are close to the mean
- b) The data points are widely spread out from the mean
- c) The dataset has more central tendency
- d) The dataset has less variability

Q6: Which measure of variability tells you the average squared deviations from the mean?

- a) Variance
- b) Range
- c) Interquartile Range (IQR)
- d) Mean

Q7: Skewness measures:

- a) The symmetry of the data distribution
- b) The peakedness of the data distribution
- c) The spread of data around the mean
- d) The central tendency of the data

Q8: What does kurtosis measure in a dataset?

- a) The skewness of the distribution
- b) The shape of the tails in a data distribution
- c) The frequency of values around the mean
- d) The number of peaks in the distribution

1.3 Frequency Distribution

Q9: Which chart type is most appropriate for displaying the frequency distribution of categorical data?

- a) Histogram
- b) Bar Chart
- c) Line Chart
- d) Pie Chart

Q10: A histogram represents:

- a) Categorical data using bars
- b) Continuous data using bars to show frequency
- c) Frequency as a percentage
- d) Data using lines to connect points

Q11: The Interquartile Range (IQR) is:

- a) The difference between the highest and lowest values in the dataset
 - b) The range of the middle 50% of the data
 - c) The square root of the variance
 - d) The distance from the mean to the highest data point
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2. Inferential Statistics

2.1 Hypothesis Testing

Q12: The Central Limit Theorem states that:

- a) The sum of the observations in any population follows a normal distribution
- b) For large sample sizes, the sampling distribution of the sample mean approaches a normal distribution, regardless of the population distribution
- c) A population will always follow a normal distribution
- d) The distribution of sample data is always skewed

Q13: A Type I error occurs when:

- a) You accept a true null hypothesis
- b) You reject a true null hypothesis
- c) You accept a false null hypothesis
- d) You reject a false null hypothesis

Q14: A two-tailed test is used when:

- a) We are only interested in deviations in one direction from the null hypothesis
- b) We are interested in deviations in both directions from the null hypothesis
- c) The distribution is skewed
- d) The sample size is small

2.2 Regression and Correlation Analysis

Q15: In regression analysis, the line of best fit is determined by minimizing:

- a) The sum of the residuals
- b) The sum of the squared residuals
- c) The total variance
- d) The correlation coefficient

Q16: Which statistic is used to measure the strength and direction of a linear relationship between two variables?

- a) Mean
- b) Standard Deviation
- c) Correlation Coefficient
- d) Median

2.3 Confidence Intervals

Q17: A 95% confidence interval means that:

- a) 95% of the sample means fall within the interval
- b) There is a 95% probability that the population mean lies within the interval
- c) 95% of the population is within the sample
- d) The sample mean is exactly at the center of the interval

Part B:

Probability Distribution of Data

Q1: Which of the following is a characteristic of a **normal distribution**?

- a) It is skewed to the right.
 - b) The mean, median, and mode are equal.
 - c) The tails never meet the horizontal axis.
 - d) It has two peaks.
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Q2: The **Poisson distribution** is best used to model:

- a) The probability of success or failure in a single trial.
 - b) The number of events occurring within a fixed interval of time or space.
 - c) Continuous random variables.
 - d) The probability of drawing a card from a deck.
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2. Central Limit Theorem (CLT)

Q3: The **Central Limit Theorem** states that the sampling distribution of the sample mean:

- a) Approaches a normal distribution as the sample size decreases.
 - b) Approaches a normal distribution as the sample size increases, regardless of the population's distribution.
 - c) Will always be skewed to the left.
 - d) Does not depend on the sample size.
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Q4: According to the **Central Limit Theorem**, for a sufficiently large sample size, the sample mean will:

- a) Always be greater than the population mean.
 - b) Have the same variance as the population.
 - c) Be approximately normally distributed.
 - d) Be equal to the sample variance.
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3. One-Tailed and Two-Tailed Tests

Q5: In a **one-tailed hypothesis test**, the critical region for rejecting the null hypothesis is:

- a) On both sides of the distribution.
 - b) On only one side of the distribution.
 - c) In the middle of the distribution.
 - d) Determined by the sample size.
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Q6: A **two-tailed test** is used when:

- a) The researcher is interested in deviations in both directions from the null hypothesis.
 - b) The researcher only cares about deviations in one direction.
 - c) The data is normally distributed.
 - d) The population mean is unknown.
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4. Type I and Type II Errors

Q7: A **Type I error** occurs when:

- a) A true null hypothesis is rejected.
 - b) A false null hypothesis is accepted.
 - c) The sample size is too small.
 - d) The population is not normally distributed.
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Q8: A **Type II error** occurs when:

- a) A false null hypothesis is rejected.
 - b) A true null hypothesis is rejected.
 - c) A false null hypothesis is accepted.
 - d) The data is normally distributed.
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Q9: Which of the following reduces the likelihood of making a **Type I error**?

- a) Increasing the sample size.
- b) Lowering the significance level (α).
- c) Increasing the significance level (α).
- d) Using a one-tailed test.

Q10: Which of the following reduces the likelihood of making a **Type II error**?

- a) Increasing the sample size.
 - b) Lowering the significance level (α).
 - c) Using a two-tailed test.
 - d) Increasing the likelihood of a Type I error.
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5. Regression and Correlation Analysis

Q11: In a **simple linear regression** model, the slope of the regression line represents:

- a) The strength of the relationship between the variables.
 - b) The predicted change in the dependent variable for a one-unit increase in the independent variable.
 - c) The correlation coefficient.
 - d) The intercept of the regression line.
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Q12: In a **correlation analysis**, a **correlation coefficient** of **-0.9** indicates:

- a) A strong negative linear relationship.
 - b) A weak positive linear relationship.
 - c) No relationship between the variables.
 - d) A moderate negative relationship.
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Q13: Which of the following best describes the **coefficient of determination (R^2)** in regression analysis?

- a) It measures the strength of the linear relationship between two variables.
 - b) It explains the proportion of variance in the dependent variable that is predictable from the independent variable.
 - c) It represents the slope of the regression line.
 - d) It is equal to the standard deviation of the residuals.
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Q14: In multiple regression analysis, **multicollinearity** occurs when:

- a) The dependent variable is correlated with the residuals.
 - b) The independent variables are highly correlated with each other.
 - c) The regression model has a low R^2 value.
 - d) There is no relationship between the dependent and independent variables.
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Q15: Which of the following statements about the **Pearson correlation coefficient** is **true**?

- a) It is always positive.
- b) It ranges from -1 to +1.
- c) A value of 0 indicates a strong correlation.
- d) It measures the relationship between two categorical variables.